

Shopping Mall Customer Data **Segmentation Analysis**

Business Analyst Project Report

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1. Problem Statement

The objective of this project is to analyze customer demographics, income levels, and spending behaviour to identify meaningful customer segments. The analysis aims to help businesses understand their customers better, improve marketing strategies, increase customer engagement, and maximize revenue through targeted campaigns.

2. Dataset Information

Total Records: 15079

Total Features: 5

The dataset was cleaned and prepared for analysis by checking for missing values, removing inconsistencies, and creating additional features such as customer segments and age groups.

The dataset contains both categorical and numerical variables which are useful for exploratory and data analysis.

3. Approach / Methodology

The following steps were performed during the analysis:

Data Cleaning

- Loaded the dataset using Python (Pandas).
- Checked for missing values and data inconsistencies.
- Verified data types and prepared the dataset for analysis.

Exploratory Data Analysis (EDA)

- Calculated average age of customers.
- Calculated average annual income.
- Analyzed spending behaviour by gender.
- Identified the age group with the highest spending score.

- Examined the relationship between annual income and spending score.

Customer Segmentation

Customers were segmented into four categories based on income and spending behaviour:

1. High Income – High Spending
2. High Income – Low Spending
3. Low Income – High Spending
4. Low Income – Low Spending

Dashboard Development

A Power BI dashboard was created containing:

- Total Customers KPI
- Average Age KPI
- Average Annual Income KPI
- Average Spending Score KPI
- Total Segments KPI
- Customer Distribution by Age Group
- Gender Distribution Analysis
- Income vs Spending Score Analysis
- Customer Segment Distribution

4. Key Insights

- The customer base consists of individuals from various age groups and income levels.
- Spending behaviour varies significantly across customer segments.
- Income alone does not determine spending behaviour.
- Some customers with lower income demonstrate high spending tendencies.
- Distinct customer groups can be targeted using segmentation strategies.

5. Business Recommendations

1. Which segment should the company target most?

The company should primarily target the High Income – High Spending segment because these customers generate the highest revenue and have strong purchasing power.

Personalized marketing campaigns and premium offerings should be focused on this segment.

2. What offers can attract low-spending customers?

To increase engagement among low-spending customers, the company can:

- Provide discounts and promotional offers.
- Introduce loyalty and reward programs.
- Offer bundle deals and seasonal campaigns.
- Send personalized product recommendations based on customer preferences.

3. How can high-income customers be retained?

High-income customers can be retained through:

- Exclusive membership programs.
- Premium products and services.
- Personalized customer support.
- Early access to new launches and special offers.
- Loyalty rewards and VIP benefits.

6. Conclusion

Customer segmentation provides valuable insights into customer behaviour and purchasing patterns. By targeting the right customer segments and implementing data-driven marketing strategies, businesses can improve customer satisfaction, increase sales, and enhance long-term profitability.